2021-Summer Paper 3 – Subspecialities of Urology

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Q1. Embryological development of urinary bladder and management of Hutch Diverticulum?

Type: LAQ (25 marks) (bladder diverticulum also answered in 2016-Summer Paper 2, Q3c)

A. Embryology of the bladder

1. Division of the cloaca (weeks 4–6)

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. The cloacal membrane perforates to form the urogenital and anal openings.
- The **Wolffian (mesonephric) duct entry** is the landmark:
  - **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
  - **Caudal (definitive) UGS** → phallic urethra in males; distal vagina and vestibule in females. The whole female urethra comes from the pelvic part.

2. Wall layers

- **Urothelium:** endoderm of the UGS.
- **Lamina propria, detrusor and adventitia:** surrounding splanchnic mesenchyme. Epithelium and mesenchyme induce each other (epithelial–mesenchymal signalling).

3. Trigone (weeks 5–6)

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the mesonephric ducts.
- Ureteric orifices migrate **cranially and laterally**; the Wolffian ducts end caudally as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric (Wolffian) tissue. Current view (mouse lineage tracing): **mostly bladder smooth muscle**, minor ureteric contribution.
- **Bud position (Weigert–Meyer):** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

4. Allantois and urachus

- **Months 4–5:** the bladder descends into the pelvis. The allantoic duct elongates and obliterates into the **urachus** (median umbilical ligament).

5. Function

- Urine production starts at **week 10**. Fetal bladder is seen from 10 weeks, almost always by 13 weeks, and empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

B. Anomalies explained by embryology

Anomaly	Embryological basis
Agenesis	Cranial UGS atrophy, or trigone not formed; survival only with ectopic ureters
Exstrophy	Cloacal membrane not reinforced by mesoderm
Duplication	Complete or incomplete; sagittal or coronal
Urachal (patent urachus, sinus, cyst, vesicourachal diverticulum)	Urachus not obliterated at different levels
Congenital diverticulum	Focal detrusor deficiency, often at the ureteric hiatus

C. Hutch diverticulum: definition

- **Paraureteral diverticulum**: mucosa herniates through the **ureteric hiatus**, between the intravesical ureter and the roof of the hiatus, superolateral to the orifice, sparing the trigone. **Hutch** divided them into primary and secondary.

	Primary (congenital)	Secondary (acquired)
Cause	Congenitally deficient detrusor at the hiatus	High pressure: PUV (posterior urethral valves), BOO (bladder outlet obstruction), neurogenic bladder (described in paraplegics), dysfunctional voiding, prior surgery
Bladder	Smooth; single, often intermittent	Trabeculated; one of many

D. Pathophysiology

- No muscle backing → **VUR**; no muscle in the sac → cannot empty → stasis, UTI (urinary tract infection), stones; tumour in adults.
- An enlarging sac damages **Waldeyer's sheath** and pulls the ureter out → UVJ (ureterovesical junction) dysfunction, reflux, later **obstruction** and renal damage.
- Large sacs compress the bladder neck → retention (vicious circle).

E. Evaluation

- **US (ultrasound):** sac, residual urine, hydronephrosis.
- **VCUG (voiding cystourethrogram), gold standard:** oblique and post-void films show size, VUR and whether the ureter enters the sac. Repeat if suspicion persists (sac is dynamic).
- **DMSA (dimercaptosuccinic acid):** scars; **MAG3 (mercaptoacetyltriglycine) renogram** if obstruction.
- **Urodynamics** if secondary; cystoscopy (orifice, neck); in adults enter the sac, cytology.

F. Management

1. Treat the cause first (secondary): valve ablation, BOO surgery, CIC (clean intermittent catheterisation) and antimuscarinics. Remodelling may make excision unnecessary.

2. Primary diverticulum:

Situation	Management
Small, asymptomatic, no VUR	Observe

Situation	Management
VUR, ureter beside the sac	As primary VUR (resolves like VUR of the same grade); many excise if it persists. Dx/HA (dextranomer/hyaluronic acid) injection possible
Ureter enters the sac	VUR will not resolve → surgery
Breakthrough febrile UTI, new scars, large sac with residual, stones, obstruction, retention	**Diverticulectomy + ipsilateral reimplantation**

3. Operative steps (open, laparoscopic or robotic):

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Cystoscopy ± ureteric catheter → cystotomy → evert sac or dissect it
from its pseudocapsule (0.5 cm mucosal cuff round the neck)
→ close detrusor defect in two layers (firm backing)
→ reimplant ureter: Cohen cross-trigonal or Politano–Leadbetter
  (tunnel 5:1) → catheter ± suprapubic tube, drain
```

- Extravesical or combined approaches suit selected cases. In adults, endoscopic outlet surgery is done first.

G. Complications and follow-up

- Ureteric injury, leak, persistent VUR, obstruction, recurrence (syndromic patients).
- US at 6–12 weeks; VCUG for febrile UTI; urodynamics if voiding is abnormal.

Also know: posterolateral multiple diverticula (Ehlers–Danlos, Williams, Menkes); hypoplasia; megacystis.

Diagram to draw: cloacal division by the urorectal septum; Hutch sac beside the ureteric orifice.

> **Key points to remember:**
> - Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
> - Hutch: herniation through the ureteric hiatus → VUR.
> - Ureter entering the sac: VUR will not resolve.
> - Secondary: treat the outlet first; else diverticulectomy + reimplantation.

Q2. Vascular Grafts in Urology.

Type: LAQ (25 marks) (SAQ version answered in 2015-Summer Paper 2, Q3e)

Assumed meaning: blood-vessel grafts and patches (conduits) used in urological surgery: transplantation, renal and IVC (inferior vena cava) tumour surgery, RPLND (retroperitoneal lymph node dissection), sarcoma, renovascular repair and autotransplantation, dialysis access and penile surgery. The chapters treat "vascular grafts" in this sense; tissue grafts (buccal mucosa) are not included.

Introduction

- A vascular graft **replaces, bypasses, extends or widens** a vessel when primary repair would leave tension or narrowing.
- **Patch**: widens a partly resected wall. **Interposition graft**: replaces a segment. **Bypass**: routes flow around a lesion.

Classification of graft materials

Type	Examples	Advantages	Disadvantages
Autologous vein	Great saphenous, gonadal, superficial femoral, donor external iliac vein	Resists infection; suits small vessels and contaminated fields	Limited length; valves; harvest wound
Autologous artery	Internal iliac (hypogastric), **Carrel aortic patch**	Best for renal artery repair	Limited supply
Allograft (homograft)	Cadaveric vein or artery	Off the shelf; avoids xenograft	Availability; degeneration
Xenograft (biological)	**Bovine or porcine pericardium**	Pliable, strong patch, low infection	Patch only
Synthetic	**PTFE (polytetrafluoroethylene, Gore-Tex)**, ringed PTFE, Dacron (polyethylene terephthalate), collagen-impregnated Dacron	Large calibre, any length	Infection needs removal; venous grafts thrombose

Principles

- **Autologous** tissue for small vessels and contaminated fields; **synthetic** for large, high-flow vessels with no bowel contamination.
- **Venous (low-pressure) grafts thrombose** more than arterial ones, so use the **widest graft** possible.
- Patch: **smooth side luminal**, rough thrombogenic side outward; edges must not invert.
- Systemic heparin before clamping; flush air and debris before tying; tension-free spatulated anastomosis with fine monofilament polypropylene (Prolene).
- Wrap synthetic grafts in **omentum** or retroperitoneal fat, away from bowel and urine.

Applications in urology

1. Renal transplantation

- **Short right renal vein**: extend with donor IVC, free donor external iliac vein, or **spiral gonadal vein graft**.
- **Multiple arteries (18–43%)**: Carrel patch; on the back table, conjoined side-to-side or end-to-side reconstruction. **Preserve the lower-pole artery** (it supplies the ureter).
- **Atherosclerotic recipient iliac artery**: mount the graft artery on a **Gore-Tex or bovine pericardial patch**.

- **Children under 15 kg**: aorta and IVC anastomosis, transperitoneal.
- **Transplant renal artery stenosis (1–23%)**: angioplasty or stent first; surgical interposition or bypass risks graft loss.

2. RCC (renal cell carcinoma) with IVC tumour thrombus

Situation	Repair
Residual lumen ≥50%	Primary closure
Residual lumen <50%	**Patch cavoplasty**: bovine or autologous pericardium, PTFE, Dacron or saphenous vein
Circumferential invasion	**Caval replacement: PTFE 16–20 mm**; contralateral renal vein reimplanted into the graft
Infrarenal bland thrombus	Filter or caval interruption, sparing lumbar collaterals

- **Leibovich–Bower patch width** = normal caval diameter × % diameter resected × 3.
- Left tumour with suprarenal caval resection: the right kidney has no collaterals, so give it new drainage (autotransplantation, or **saphenous vein graft** to the splenic, portal or inferior mesenteric vein).
- After caval replacement: low-dose heparin, then **lifelong aspirin 81 mg**.

3. PC-RPLND (post-chemotherapy RPLND)

- Vascular procedures in **9–15%**; aorta most often.
- **IVC**: narrowing <25% → primary repair; >25% → patch or interposition graft. **Ligation is tolerated** (slow occlusion builds collaterals).
- **Aorta**: never ligated. Small injury → repair or pericardial patch; resection → **PTFE or Dacron** (about 1% need replacement). Graft also if the duodenum is injured, to prevent aortoduodenal fistula.

4. Retroperitoneal sarcoma

- Caval resection ± reconstruction (Dacron, PTFE, homograft); grafts **thrombose in 10–40%**; anticoagulate at least 3 months.

5. Renovascular repair and autotransplantation

- Bench (ex vivo) repair of branch aneurysm or stenosis with internal iliac artery or saphenous vein, then autotransplantation (see Q3e). Traumatic renal artery repair: attempt only for solitary kidney or bilateral injury (warm ischaemia <120 minutes).

6. Dialysis access

- **AV (arteriovenous) graft (PTFE)** when no native fistula is possible: usable in 2–3 weeks; more stenosis, thrombosis and infection than AVF (arteriovenous fistula).

7. Iatrogenic vascular injury

- SMA (superior mesenteric artery): repair or graft bypass. SMV (superior mesenteric vein): autologous vein graft; Gore-Tex only if no vein.

8. Penile surgery

- **Grayhack** saphenous vein–cavernosal shunt (priapism). Vein grafts for Peyronie tunical defects (survive by endothelial perfusion); synthetic grafts not recommended.

Complications

- **Thrombosis**, anastomotic stenosis (intimal hyperplasia), bleeding, pseudoaneurysm.
- **Graft infection** (synthetic → usually removal); graft–enteric fistula.
- **Ureteric**: transient hydronephrosis after aorto-iliac grafts (12–20%), symptomatic stricture 1–2%, **ureteroarterial fistula** (treat with endovascular covered stent).

Recent advances

- Endovascular covered stent grafts; bioengineered acellular vessels (under study); robotic autotransplantation.

Diagram to draw: IVC patch cavoplasty and PTFE caval replacement with left renal vein reimplanted.

Key points to remember:

- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- IVC: lumen <50% → patch; invaded wall → PTFE 16–20 mm.
- PC-RPLND: IVC may be ligated; aorta must be grafted.
- Transplant: Carrel patch; preserve the lower-pole artery.

Q3a. Koyanagi Repair: Indication, procedure and complications?

Type: SAQ (10 marks) (proximal hypospadias also answered in 2017-Winter Paper 3, Q1)

Definition

- **One-stage repair of proximal hypospadias** with a **meatus-based parameatal foreskin flap** (Koyanagi). Skin strips round the meatus continue into the inner prepuce on both sides; they are swung ventrally and tubularised.
- **Principle:** the neourethra stays attached to the native meatus, so there is **no circular proximal anastomosis**, and the urethral plate can be divided to correct severe chordee.

Indications

- **Proximal hypospadias** (penoscrotal, scrotal, perineal) with **severe curvature** needing plate transection, and an **ample, unoperated prepuce**.
- **Contraindications:** prior circumcision or failed repair; small glans (give preoperative testosterone); curvature needing a ventral corporal graft (stage instead).

Procedure

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Stay suture; artificial erection
→ incise round the meatus, along both sides of the plate to the
  corona and round the inner prepuce (meatus-based "wings")
→ deglove; divide the plate; excise dysplastic dartos
→ residual curvature <30° → dorsal midline plication
→ split the dorsal prepuce; bring both wings ventrally on dartos pedicles
→ join inner edges, then outer edges, over a catheter → tube
→ glanuloplasty, meatus at the tip
→ dartos barrier layer; skin cover ± scrotoplasty
→ stent 7–10 days; compressive dressing
```

- Modified and "neo-modified" versions add flap fixation and barrier layers to reduce fistula and diverticulum.

Complications

Complication	Cause
Fistula (commonest)	Long suture line; repair ≥6 months later
Meatal stenosis, stricture	Distal flap ischaemia
Diverticulum	No spongiosum; flap not fixed to the shaft
Glans dehiscence	Tension
Residual or recurrent curvature	Plication alone in severe cases
Flap necrosis, infection, poor cosmesis	Pedicle injury

- Proximal repairs overall have **23–68%** complications, so many now prefer **two-stage repair** (STAG (staged tubularised autograft), Bracka).

Also know: Duckett TPIF (transverse preputial island flap); onlay island flap; Byars two-stage.

Diagram to draw: meatus-based foreskin wings split dorsally and joined ventrally into a tube.

> **Key points to remember:**
> - One stage; meatus-based parameatal foreskin flap.
> - Plate divided; no proximal circular anastomosis.
> - Fistula commonest; diverticulum, meatal stenosis.

Q3b. Antenatal management of Fetal Uropathy.

Type: SAQ (10 marks)

Introduction

- Mostly **UTD (urinary tract dilation)** on the 20-week scan, usually mild and transient. The serious group is **LUTO (lower urinary tract obstruction)**: PUV (posterior urethral valves), urethral atresia, PBS (prune-belly syndrome).
- After **16 weeks amniotic fluid is fetal urine**; oligohydramnios causes **pulmonary hypoplasia**.

Risk stratification (UTD consensus, 2014)

Parameter	A1 (low risk)	A2–3 (increased risk: any one)
APD (anteroposterior pelvic diameter) 16–27 weeks	4 to <7 mm	≥7 mm
APD ≥28 weeks	7 to <10 mm	≥10 mm
Calyces	Central or none	Peripheral
Parenchyma, ureter, bladder	Normal	Abnormal
Amniotic fluid	Normal	Unexplained oligohydramnios

- **A1:** repeat US (ultrasound) after 32 weeks. **A2–3:** US every 4 weeks to delivery. Both: postnatal US after 48 hours.

Evaluation of suspected LUTO

- Male fetus, thick-walled "keyhole" bladder, bilateral hydroureteronephrosis, oligohydramnios.
- Anomaly scan, **fetal echocardiogram, karyotype** (one-fifth of megacystis is aneuploid); fetal MRI (magnetic resonance imaging) if unclear.
- **Serial fetal urine** (3 samples, 48–72 hours apart). Favourable: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <6 mg/L; not validated.
- **Ruano staging:** I mild, normal fluid → observe. **II severe, function preserved** (oligohydramnios, favourable urine) → intervene. III cysts, dysplasia, unfavourable urine → for lungs only.

Options

Option	Notes
Observation, serial scans	Most cases
VAS (vesicoamniotic shunt)	Dislodgement, blockage, preterm labour
Fetal cystoscopy + laser valve ablation	Diagnoses and treats; fistula up to 10%
Serial amnioinfusion	Lungs in anhydramnios; under trial
Early delivery / termination	Mature lungs, falling fluid / poor prognosis

- **PLUTO (2013):** shunt vs conservative (31 fetuses): better perinatal survival, **no renal benefit**.

Counselling and delivery

- Multidisciplinary team: fetal medicine, paediatric urology, nephrology, neonatology.
- Intervention improves survival, not renal function; complications **over 40%**.
- Deliver where the bladder can be decompressed at once; prophylaxis, then VCUG (voiding cystourethrogram).

Diagram to draw: keyhole bladder with bilateral hydroureteronephrosis and shunt.

Key points to remember:

- A1: one scan after 32 weeks; A2–3: every 4 weeks.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

Q3c. Post traumatic bladder neck injury: acute and delayed management?

Type: SAQ (10 marks)

Introduction

- The bladder neck is the **internal sphincter**. It is torn mainly in **pelvic fracture**, often with prostatic urethral, rectal or vaginal injury; **commoner in children**.
- Missed injury causes **incontinence, fistula and stricture**.

Diagnosis

- Pelvic fracture with gross haematuria, retention or blood at the meatus; **vaginal examination** in females.
- **RUG (retrograde urethrogram)**, then **retrograde CT or plain cystography** (350 mL).
- Grading: AAST (American Association for the Surgery of Trauma) bladder **grade V** = laceration into the bladder neck or ureteric orifice; Goldman type IV = bladder rupture extending into the proximal urethra.

Acute management

- Resuscitate; if unstable, damage control (drainage), repair later.
- **Bladder neck injury needs immediate repair** (AUA (American Urological Association) and EAU (European Association of Urology)), unlike isolated posterior urethral distraction (suprapubic tube, delayed repair).

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Lower midline incision → anterior cystotomy
→ identify ureteric orifices (stent if near)
→ minimal debridement; re-approximate neck to prostatic urethra,
  layered absorbable sutures over a 22 Fr catheter
→ rectal or vaginal tear: separate layers, no overlapping suture
  lines, interpose omentum (colostomy for rectum)
→ remove bone spicules; drain ± suprapubic tube
```

- Catheter 2–3 weeks; cystogram before removal.

Delayed management (3–6 months)

Problem	Management
Associated urethral distraction	Cystogram + RUG + endoscopy; **anastomotic urethroplasty first**, not combined with bladder neck surgery
Bladder neck contracture	Endoscopic incision or resection; open Y-V plasty if refractory
Incontinence (both sphincters damaged)	Urodynamics; **AUS (artificial urinary sphincter)**; bladder neck reconstruction (Young–Dees–Leadbetter) or fascial sling; last resort **bladder neck closure + Mitrofanoff** ± augmentation
Fistula	Layered repair with tissue interposition

- Bladder neck appearance on imaging **does not predict** continence; restore continuity first, then treat incontinence.
- Continence is poor whenever the bladder neck is involved (paediatric series: incontinence up to 20% overall).

Diagram to draw: pelvic fracture with bladder neck tear and repair over a catheter.

Key points to remember:

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

Q3d. Management of AML.

Type: SAQ (10 marks)

Assumed meaning: AML = renal angiomyolipoma (not acute myeloid leukaemia).

Introduction

- Benign **PEComa (perivascular epithelioid cell tumour)** of dysmorphic **vessels, smooth muscle and fat**. Female predominance; fourth–fifth decades.
- **Sporadic** (solitary, small) or with **TSC (tuberous sclerosis complex**; AML in 55–90%: younger, bilateral, multifocal, larger) or LAM (lymphangioleiomyomatosis).
- **Mechanism:** loss of **TSC2** (tuberin, 16p13) or TSC1 (hamartin, 9q34) → unchecked **mTOR (mammalian target of rapamycin)** → growth and angiogenesis.
- **Epithelioid AML**: little fat; aggressive (venous extension, metastasis) → treat as RCC (renal cell carcinoma).

Clinical features

- Mostly incidental. Large tumours bleed: **Wunderlich syndrome** (spontaneous retroperitoneal haemorrhage; Lenk triad: flank pain, mass, shock). CKD (chronic kidney disease) in TSC.

Diagnosis

- **CT (computed tomography): macroscopic fat (about −15 to −20 HU) on non-contrast images is diagnostic.**
- US (ultrasound) hyperechoic (so are some RCCs); MRI (magnetic resonance imaging): fat-suppression signal loss, India-ink artefact.
- Fat-poor lesion → MRI ± biopsy (HMB-45 (human melanoma black-45), Melan-A, smooth muscle actin positive).
- **Aneurysm ≥5 mm** on CT angiography predicts bleeding.

Management algorithm (Restrepo 2022)

Situation	Management
Acute haemorrhage	Resuscitate → **SAE (selective arterial embolisation)** → nephrectomy if SAE fails or unavailable
Symptomatic (pain, haematuria) or poor access to care	SAE, NSS (nephron-sparing surgery), ablation in selected cases
Asymptomatic sporadic with **>6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnancy**	Active treatment
None of these	**Surveillance**: US or CT at 6 and 12 months, then yearly
TSC, asymptomatic, **>3 cm**	**Everolimus**; MRI every 1–3 years

- The old **4 cm** cut-off overtreats; EAU (European Association of Urology) 2023 favours surveillance for most.

Treatments

- **SAE:** first line; **post-embolisation syndrome** (fever, pain, leucocytosis) common; abscess, infarction.
- **Partial nephrectomy:** preferred surgery, essential in TSC (multifocal, recurrent).

- **Thermal ablation** (radiofrequency, cryo): limited data.
- **Everolimus (EXIST-2):** ≥50% shrinkage in **42% vs 0%** (placebo); stomatitis, acne, hyperlipidaemia.

Diagram to draw: management algorithm by bleeding, symptoms, size and TSC.

> **Key points to remember:**
>
> - Macroscopic fat on non-contrast CT is diagnostic.
> - Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
> - SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Q3e. Role of auto transplant of kidney in Urology.

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 3, Q3c)

Definition

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** if needed. First done in **1963** for proximal ureteric injury.
- **Principle:** bringing the kidney to the bladder replaces the ureter without bowel; the bench gives a cold, bloodless field.

Role (indications)

Group	Examples
Ureteric loss (commonest)	Long iatrogenic or traumatic loss; failed repairs (**delayed, never acute**)
RPF (retroperitoneal fibrosis)	When ureterolysis is impossible and function is satisfactory
Tumour	Total ureterectomy for UTUC (upper tract urothelial carcinoma), solitary kidney; bench partial nephrectomy; left RCC (renal cell carcinoma) needing suprarenal caval resection (new drainage for the right kidney)
Vascular	Complex branch renal artery aneurysm, stenosis or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker syndrome
Pain	Selected loin pain–haematuria syndrome

- **Chosen over ileal ureter** when the other kidney is absent or poor, or bowel is unsuitable (creatinine >2 mg/dL, inflammatory bowel disease, radiation enteritis).
- **Contraindications:** poor function, severe iliac atherosclerosis, unfit.

Procedure

```
CT (computed tomography) angiography, renogram
→ harvest as for live donation (open, laparoscopic, robotic):
  maximal vessel length, preserve ureteric blood supply
→ cold perfusion ± bench repair
→ Gibson incision, extraperitoneal iliac fossa
→ vein to external iliac vein; artery to external iliac
  (end-to-side) or internal iliac (end-to-end)
→ pyelovesicostomy (± Boari flap) or ureteroneocystostomy
→ stent, catheter, drain
```

- Laparoscopic harvest: faster recovery; **robotic autotransplantation** emerging.

Complications and outcome

- Thrombosis, bleeding, urine leak, stricture, lymphocele, infection.
- **Renal unit loss 5–10%**; major (Clavien–Dindo ≥III) complications about 15–20%.
- **Long-term success about 90%.** Doppler ultrasound early; renogram later.

Also know: alternatives: Boari flap, TUU (transureteroureterostomy), ileal ureter, buccal ureteroplasty, renal descensus.

Diagram to draw: kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

Key points to remember:

- For long ureteric loss when the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Unit loss 5–10%; success about 90%.

Q3f. Urinary biomarkers in diagnostic evaluation of hematuria.

Type: SAQ (10 marks)

Introduction

- Haematuria is evaluated by **cystoscopy and upper tract imaging**; urine tests are non-invasive **adjuncts** for bladder cancer.
- Dysmorphic RBCs (red blood cells), casts or proteinuria suggest a glomerular source → nephrology, plus urological evaluation.

Urine cytology

- **Sensitivity 31–62%, specificity 94–100%**: good for **high-grade tumours and CIS (carcinoma in situ)**, poor for low grade.
- **Paris System:** negative for HGUC (high-grade urothelial carcinoma), atypical, suspicious, HGUC (by nuclear-to-cytoplasmic ratio and nuclear features).
- Positive cytology, normal cystoscopy → random and prostatic urethral biopsies, upper tract washings.

Urinary biomarkers (sensitivity/specificity, Chou 2015 meta-analysis)

Marker	Principle	Sens / spec	Note
NMP22 (nuclear matrix protein 22)	Nuclear protein; lab or point-of-care	0.69/0.77	False positives: inflammation, stones
BTA (bladder tumour antigen) stat / TRAK	Complement factor H–related protein; dipstick or ELISA (enzyme-linked immunosorbent assay)	0.64/0.77	**False positive with blood** (factor H in serum)
UroVysion FISH (fluorescence in situ hybridisation)	Gain of chromosomes 3, 7, 17; loss of 9p21	0.63/0.87	After atypical cytology
ImmunoCyt	Antibodies to CEA (carcinoembryonic antigen) + two mucins	0.78/0.78	Highest sensitivity; operator dependent
Cxbladder	RT-PCR (reverse transcription polymerase chain reaction), 5 mRNAs	about 0.82/0.85	Needs validation

- **Also know:** Xpert Bladder (mRNA); AssureMDx (FGFR3, TERT, HRAS mutations + methylation).
- **Pattern:** more sensitive, less specific than cytology; miss small low-grade tumours.

Role in haematuria (guidelines)

- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020: no cytology or markers in the initial evaluation of microhaematuria**; cytology only after a negative work-up if CIS is suspected.
- **Gross haematuria:** cytology with cystoscopy and CT (computed tomography) urography. A negative marker still leaves **≥10% cancer probability**, so cystoscopy is still needed.
- **No marker replaces cystoscopy** (AUA, EAU (European Association of Urology)); a negative test may one day help triage low-risk patients.
- **Recent advances:** urinary tumour DNA; methylation assays (Bladder EpiCheck).

Diagram to draw: haematuria pathway with markers as adjuncts.

Key points to remember:

- Cytology: specific for high grade/CIS, insensitive for low grade.
- BTA is falsely positive with blood.
- No markers in initial microhaematuria work-up; none replaces cystoscopy.

Rapid Revision Sheet – 2021-Summer Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Embryological development of urinary bladder and management of Hutch Diverticulum? (LAQ)

- Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
- Hutch: herniation through the ureteric hiatus → VUR.
- Ureter entering the sac: VUR will not resolve.
- Secondary: treat the outlet first; else diverticulectomy + reimplantation.

Q2. Vascular Grafts in Urology. (LAQ)

- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- IVC: lumen <50% → patch; invaded wall → PTFE 16–20 mm.
- PC-RPLND: IVC may be ligated; aorta must be grafted.
- Transplant: Carrel patch; preserve the lower-pole artery.

Q3a. Koyanagi Repair: Indication, procedure and complications? (SAQ)

- One stage; meatus-based parameatal foreskin flap.
- Plate divided; no proximal circular anastomosis.
- Fistula commonest; diverticulum, meatal stenosis.

Q3b. Antenatal management of Fetal Uropathy. (SAQ)

- A1: one scan after 32 weeks; A2–3: every 4 weeks.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

Q3c. Post traumatic bladder neck injury: acute and delayed management? (SAQ)

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

Q3d. Management of AML. (SAQ)

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Q3e. Role of auto transplant of kidney in Urology. (SAQ)

- For long ureteric loss when the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Unit loss 5–10%; success about 90%.

Q3f. Urinary biomarkers in diagnostic evaluation of hematuria. (SAQ)

- Cytology: specific for high grade/CIS, insensitive for low grade.
- BTA is falsely positive with blood.
- No markers in initial microhaematuria work-up; none replaces cystoscopy.